YouTube Trending Video Analysis and View Prediction Using Machine Learning

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1. Introduction

YouTube is one of the most popular video-sharing platforms where millions of videos are uploaded daily. Only a small number of these videos appear in the trending section. Understanding what makes a video trending can help content creators and analysts understand viewer engagement.

This project analyzes trending YouTube video statistics and identifies patterns in views, likes, comments, and channel performance. A machine learning model is also used to predict video views based on engagement metrics.

2. Dataset Description

The dataset used in this project is **Trending YouTube Video Statistics**. The **INvideos.csv** dataset was used, which contains **37,352 records and 16 attributes**.

Important columns include:

- Video Title
- Channel Title
- Category ID
- Views
- Likes
- Dislikes
- Comment Count

These features help analyze video popularity and viewer engagement.

```
[5 rows x 16 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 37352 entries, 0 to 37351
Data columns (total 16 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   video_id                37352 non-null  object
 1   trending_date           37352 non-null  object
 2   title                   37352 non-null  object
 3   channel_title           37352 non-null  object
 4   category_id             37352 non-null  int64
 5   publish_time            37352 non-null  object
 6   tags                    37352 non-null  object
 7   views                   37352 non-null  int64
 8   likes                   37352 non-null  int64
 9   dislikes                37352 non-null  int64
 10  comment_count           37352 non-null  int64
 11  thumbnail_link          37352 non-null  object
 12  comments_disabled       37352 non-null  bool
 13  ratings_disabled        37352 non-null  bool
 14  video_error_or_removed  37352 non-null  bool
 15  description             36791 non-null  object
dtypes: bool(3), int64(5), object(8)
memory usage: 3.8+ MB
None
```

dataset preview and dataset information

3. Tools and Technologies

The project was implemented using the following tools:

- **Python** – Programming language
- **Pandas** – Data analysis
- **Matplotlib & Seaborn** – Data visualization
- **Scikit-learn** – Machine learning model
- **Visual Studio Code** – Development environment

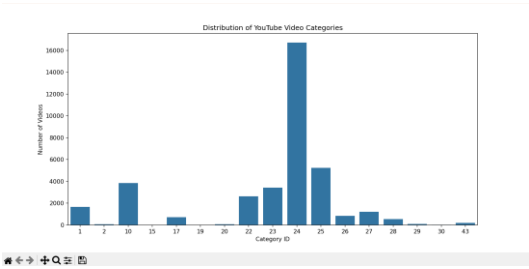
4. Data Analysis

Different analyses were performed to understand trending video patterns.

- Top 10 most viewed videos were identified.
- Channels with the highest total views were analyzed.

- Videos with the highest likes were examined.

The results showed that large entertainment channels such as **T-Series and Marvel Entertainment** receive the highest number of views.



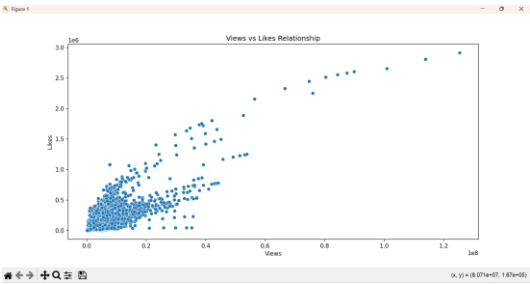
(Top Viewed Videos / Top Channels)

5. Data Visualization

Visualizations were created to understand the relationship between engagement metrics.

A **Views vs Likes scatter plot** showed a strong positive relationship between the number of views and likes.

A **correlation heatmap** was also generated to analyze relationships between views, likes, dislikes, and comments.



scatter plot and correlation heatmap.

6. Machine Learning Model

A **Linear Regression model** was used to predict video views based on engagement metrics.

Input features used:

- Likes
- Dislikes
- Comment Count

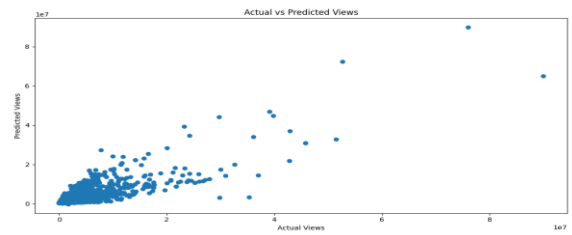
The model was trained using training data and tested using testing data.

Model Performance:

R^2 Score: 0.768

- **Mean Absolute Error:** 624004

These results indicate that the model can reasonably predict video views based on engagement data.



(prediction results and accuracy)

7. Conclusion

The analysis of YouTube trending videos shows that likes, comments, and audience engagement strongly influence video popularity. Channels producing consistent and engaging content appear more frequently in the trending list.

The machine learning model demonstrated that engagement metrics can be used to predict video views with reasonable accuracy. This project highlights how data analysis and machine learning can help understand online video trends.